BRICS — Technical Scope of Work (SOW) & SRS

Step 3: Technical Requirement Specification & Feature Scope | TVAM KEY SOFTWARE PRIVATE LIMITED

1. Technical Architecture & Stack

- **Mobile Supervisor App**: Flutter / Dart (Cross-platform iOS & Android) — Offline SQLite sync engine.
- **Web Admin Panel**: Next.js 14 (App Router) + React 18 + TypeScript + Tailwind CSS.
- **Backend REST API**: Node.js (NestJS / Express) + TypeORM.
- **Database Engine**: PostgreSQL (Multi-tenant relational schema).
- **Cloud Infrastructure**: Render / Vercel with Supabase PostgreSQL.

2. Feature Requirement Matrix (Phase 1 MVP vs Phase 2)

Module #	Module Name	Scope & Feature Description	Phase 1 (MVP)	Phase 2
1	Auth & RBAC	Multi-company, multi-kiln profile setup, financial year config, role permissions	Included	—
2	Party Masters	Directories & real-time ledgers for Customers, Suppliers, Sardars, Transporters	Included	—
3	Production	Physical lifecycle: Soil -> Brick Making -> Drying -> Firing -> Sorting -> Stock	Included	—
4	Labour Engine	Attendance tracking, cash advances, piece-rate wages per 1,000 bricks, settlements	Included	—
5	Inventory	Stock for Raw Materials (Coal, Soil) and Finished Bricks by grade	Included	—
6	Vehicle Transport	Trip logs, diesel consumption, loading/unloading fees, transport profitability	Included	—
7	Purchases	PO -> GRN -> Supplier Bill -> Stock Update -> Supplier Payment	Included	—
8	Sales & Invoicing	Quotations -> Sales Orders -> Tax Invoice (GST) -> Delivery Challan	Included	—
9	Dispatch	Links invoice, truck, grade, loading group. Auto-deducts finished inventory	Included	—
10	Accounting	Chart of Accounts, Day Book, General Ledger, Trial Balance, P&L;, Balance Sheet	Included	—

11	GST Engine	CGST/SGST/IGST calculations, HSN mapping, GSTR-1 & GSTR-3B registers	Included	—
12	Supervisor App	Field mobile app for supervisors to log production, attendance, dispatches	Included	—
13	AI Anomaly	Predictive coal efficiency & anomaly alerts	—	Phase 2
14	Hardware IoT	Weighbridge scale & kiln thermal sensor integration	—	Phase 2

3. Explicit Exclusions ("What is NOT Included")

- **Third-Party Fees**: SMS/WhatsApp gateway charges (paid directly to MSG91/Twilio), Apple ($99/yr) & Play Store ($25) dev fees.
- **Hardware Equipment**: Weighbridge scales, thermal cameras, PCs, printers, or mobile handsets.
- **Historical Data Entry**: Manual data entry of previous years paper records prior to kickoff.